Advanced Financial Modeling, Module 9: Advanced LBO Model (KKR / Viridor)

In this module, you'll complete a 4-hour case study based on KKR's leveraged buyout of Viridor, a waste management company in the U.K. It is based on a full 3-statement model and includes more advanced features, such as stub periods, a dividend recap, a Working Capital target, Revolver Fees, OID, and IRR and recovery analyses for lenders.

There are also full investment recommendation presentations for both the equity investors and the lenders at the end.

AFM-09-01: Advanced LBO Model – Overview and Data Gathering (14:39)

In this first lesson, you'll get an introduction to the KKR / Viridor case study, and you'll learn about the deal parameters and where to find the required data.

Table of Contents:

- **2:01:** Deal Overview
- **7:16:** Data Gathering
- **13:35:** Recap and Summary

Quiz Questions:

13:35 Why is it very difficult to find the proper data for an LBO of a divested company?

a) The parent company might not disclose full, historical financials.
b) It may not be clear how to split revenue and expenses between the Parent and Sub Co.
c) Automated sources such as Capital IQ or FactSet might also be rough estimates.
d) All of the above.

Explanation: Everything above is true. These all represent reasons why it's often difficult to find data for divestitures unless the parent company explicitly discloses all the information in its filings.

AFM-09-02: Viridor – Revenue Projections (10:37)

You'll learn how to project Viridor's revenue from existing facilities and newer ERF plants across different scenarios in this lesson.

Table of Contents:

- **1:28:** Existing Facilities and ERF Plants

- **2:48:** Waste Throughput
- **6:25:** Pricing Growth
- **8:47:** Revenue in Different Scenarios
- **9:51:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-03: Viridor – Expense Assumptions (11:16)

You'll learn how to forecast Viridor's expense drivers in this lesson, including its employee count, employee expenses, raw material costs, and tax rate.

Table of Contents:

- **0:30:** Historical Expenses
- **5:36:** Expense Drivers
- **10:01:** Recap and Summary

Quiz Questions:

10:00 If there's no specific guidance provided for a company's expenses, what's the safest way to forecast something like employee costs in a financial model?

a) Make them a percentage of revenue.
b) Assume they grow in-line with the employee count and wages that rise in-line with inflation.
c) Set them such that the company's margins stay in a similar range in the projected period.
d) Any of the above could work.

Explanation: Anything here could work, depending on the numbers and historical trends. These are all common approaches to forecasting expenses so that the company's margins stay in a similar range. The main drawback is that if the company's margins are expected to change for some reason, these methods will be less effective.

AFM-09-04: Viridor – CapEx, Depreciation, and Leases (12:44)

In this lesson, you'll project both Growth and Maintenance CapEx for Viridor, set up a Depreciation schedule based on the construction times and useful lives, and make simple assumptions for the leases on the company's non-ERF facilities.

Table of Contents:

- **0:55:** Growth CapEx and Depreciation
- **7:23:** Maintenance CapEx and Leases
- **11:37:** Recap and Summary

Quiz Questions:

11:37 In a Depreciation schedule in which each asset takes multiple years to build, why can't the Depreciation start in the same year as the initial CapEx spent on the asset?

a) **Depreciation can begin only once an asset is finished and put into use.**
b) It's not accurate to do this because the CapEx in Year 1 won't be fully depreciated in this case.
c) This doesn't make sense because the CapEx is typically incurred on the final day of the year.
d) All of the above.

Explanation: The first answer is correct. The short, simple answer is that a company can start depreciating an asset only after the asset is finished and put into use. The second answer is factually incorrect since all CapEx related to a new asset will eventually be depreciated, and the third answer is not a good choice because there are many ways around this issue; for example, you could use the mid-year convention and assume that CapEx is incurred, on average, halfway through each year, in which case there could be CapEx and Depreciation on the same asset in the same year.

AFM-09-05: Viridor – Balance Sheet and Cash Flow Statement Drivers (10:21)

You will set up the Balance Sheet and Cash Flow Statement assumptions for Viridor in this lesson, including the Minimum Cash and Debt Issuance formulas.

Table of Contents:

- **0:42:** Historical Percentages
- **3:41:** Projections
- **9:18:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-06: Viridor – Projecting and Linking the Statements (15:05)

In this lesson, you'll project and link Viridor's financial statements over the 6-year holding period, and you'll see how the Cash balance ends at very different numbers in the different scenarios.

Table of Contents:

- **0:39:** Income Statement
- **2:58:** Balance Sheet, Part 1
- **4:29:** Cash Flow Statement
- **9:32:** Balance Sheet, Part 2
- **13:54:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-07: Transaction Assumptions and Sources & Uses for a Cash-Free, Debt-Free Deal with a Working Capital Adjustment (14:51)

You'll set up the LBO transaction assumptions in this lesson, including the Purchase Enterprise Value and Equity Value, the Sources & Uses schedule, and the adjustments required for a cash-free/debt-free divestiture deal with a Working Capital adjustment at deal close.

Table of Contents:

- **1:56:** Transaction Assumptions
- **5:51:** Debt, OID, and Issuance Fees
- **7:56:** Sources & Uses
- **13:25:** Recap and Summary

Quiz Questions:

13:25 Suppose that a company's Working Capital at deal close is $200 vs. a targeted level of $150. How does this change the total amount the *PE firm* pays and the total amount that *goes to the shareholders*?

a) The PE firm pays $50 extra, and the selling shareholders get $50 extra.
b) The PE firm pays the same amount, and the selling shareholders get $50 less.
c) The PE firm pays $50 less, and the selling shareholders get $50 more.
d) The PE firm pays the same amount, and the selling shareholders get $50 more.

Explanation: The last answer is correct. When there's a Working Capital adjustment, the Purchase Enterprise Value is adjusted up or down, and the PE firm either "takes" some of the company's Working Capital or funds additional Working Capital such that the effective price the PE firm remains the same. So, in this case, the Purchase TEV would increase by $50, but the PE firm would also "take" $50 of the company's Working Capital.

This adjustment mostly affects the selling shareholders because it means they will get more or less depending on how they manage the company's finances before the deal closes. If there's a WC surplus, they'll get this surplus amount because it increases the Purchase TEV and Purchase Equity Value; if there's a WC deficit, this deficit gets deducted from both the Purchase TEV and Purchase Equity Value, reducing the proceeds to the shareholders.

AFM-09-08: Stub Period Support on the Income Statement and Cash Flow Statement (13:42)

In this lesson, you'll set up support for the stub periods on the financial statements so that the leveraged buyout can close midway through the company's next fiscal year.

LESSON NOTE: From 10:20 to 10:30, we "forget" to copy across the links at the top of the CFS. We return and fix this issue starting at 13:00 in the Recap and Summary section.

Table of Contents:

- **1:21:** Income Statement
- **7:10:** Cash Flow Statement
- **12:19:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-09: Balance Sheet "Rollforward" to Deal Close Date (10:31)

You'll link in Viridor's historical Balance Sheets and "roll forward" the first projected one to the transaction close date in this lesson; you'll then use this modified version in the transaction assumptions and Working Capital adjustments.

Table of Contents:

- **1:03:** Historical Balance Sheets
- **2:13:** Working Capital Line Items
- **3:39:** Other Balance Sheet Line Items
- **7:11:** Transaction Adjustments
- **9:22:** Recap and Summary

Quiz Questions:

9:21 What's the best way to "roll forward" the Balance Sheet when a transaction closes in between two fiscal years?

a) Link in the stub period CFS numbers.
b) Multiply line items by the stub period fraction.
c) The best method depends on the specific line item.

Explanation: The last answer choice is correct. The "best method" to roll forward the Balance Sheet up through the transaction close date depends on the line item. With the Working Capital line items, it's easiest to multiply by fractions to estimate their values as of the close date. But with items such as Net PP&E, Other Intangibles, and Total Debt, it's easiest to link to the corresponding Cash Flow Statement line items from the stub period there.

AFM-09-10: Purchase Price Allocation and Balance Sheet Adjustments (10:42)

In this lesson, you'll set up the Purchase Price Allocation schedule and the Balance Sheet adjustments for this deal, including changes due to its status as a cash-free, debt-free deal and the Working Capital adjustments.

Table of Contents:

- **0:58:** Purchase Price Allocation
- **3:41:** Balance Sheet Adjustments
- **9:17:** Recap and Summary

Quiz Questions:

9:16 Which of the following Balance Sheet adjustments occurs DIRECTLY because a deal is structured as an asset purchase rather than a stock purchase?

a) The company's existing Deferred Tax Liability (DTL) is written down.
b) No new DTL is created since assets are written up for both book and tax purposes.
c) The company's Cash balance immediately post-transaction falls to its Minimum Cash.
d) All of the above.

Explanation: The second answer choice is the best one. In an asset purchase, assets are written up for both book and tax purposes, so there's no book vs. cash tax difference after the deal closes, which means no Deferred Tax Liability is created. The first answer choice is true in most deals of all types, and the third answer has nothing to do with asset vs. stock purchases but is instead related to the cash-free/debt-free structure.

AFM-09-11: Transaction Effects and Projecting and Linking the Post-Transaction Balance Sheet (14:19)

You'll calculate transaction effects, such as the amortization of new line items, and then you'll use this information to link and project the Balance Sheet after the deal closes, from the stub period through the end of the holding period.

Table of Contents:

- **0:52:** Income Statement
- **5:58:** Cash Flow Statement
- **7:10:** Balance Sheet
- **12:48:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-12: Debt Schedule, Part 1: Revolver and Mandatory and Optional Repayments (13:38)

In this lesson, you'll set up the first part of the Debt Schedule, including the Cash Flow Available for Debt Repayment, the Revolver, and the mandatory and optional Debt principal repayments.

LESSON NOTE: At 3:00, we forget to add the PIK Interest to the face values for the last two debt tranches (Subordinated Notes and Mezzanine); we return and fix this issue in the next lesson when checking the Debt Schedule.

Table of Contents:

- **1:11:** Mandatory Repayments
- **4:55:** Cash Flow Available and Revolver
- **7:13:** Optional Repayments
- **11:17:** Linking the Debt Repayments
- **12:30:** Recap and Summary

Quiz Questions:

12:30 In the Optional Debt Repayment schedule of an LBO, why do we not need to add the Revolver Borrowing to the remaining Revolver balance in the period when comparing the Revolver balance to the cash flow available * sweep percentage?

a) **Because if Revolver Borrowing is required, there cannot be any optional repayments.**
b) Because the Revolver never has mandatory repayments – only optional ones.
c) Because 100% of the company's available cash flow is always used to repay the Revolver first.
d) All of the above.

Explanation: The first answer choice is correct. When a Revolver draw is required, the company cannot possibly make any optional Debt principal repayments because it means the company's Cash balance would end below its Minimum Cash – which is why we need to draw on the Revolver for additional funds. Therefore, it is unnecessary to add the Revolver Borrowing when checking his case. The fact that the Revolver has no mandatory repayments has nothing to do with this question, so the second answer is wrong. And the 100% cash flow sweep for the Revolver also has nothing to do with this question because even with the highest seniority and a 100% sweep, any tranche of Debt – except for the Revolver – could potentially have additional draws that we need to add to the "current balance" before making optional repayments.

AFM-09-13: Debt Schedule, Part 2: Net Interest Expense and Revolver Fees (15:52)

You'll calculate the Net Interest Expense and the Revolver Commitment and Overdraw Fees in this lesson, including support for both average Debt balances (circular references) and beginning Debt balances (no circular references).

Table of Contents:

- **0:51:** Net Interest Expense
- **9:08:** Revolver Fees
- **11:46:** Linking the Statements
- **14:48:** Recap and Summary

Quiz Questions:

14:48 What is the MAIN effect of the Commitment Fees on a partially undrawn Revolver along with the Overdrawn Fees?

a) It depends on how much of the Revolver is drawn.
b) They raise the effective interest rate.
c) They incentivize the company to draw on as much of the Revolver as possible.
d) They encourage the company to draw on other sources of Debt before the Revolver.

Explanation: The second answer choice is the best one. Since the fees apply to the Revolver regardless of the percentage drawn, they raise the effective interest rate. The first answer is a true statement, but it doesn't answer the question – yes, the exact effect depends on the drawn percentages, but even without knowing that, we can say that the effective interest rate goes up. These fees do not incentivize the company to draw on the maximum Revolver because the cash interest expense would also increase. And they don't encourage the company to use other sources of Debt because the Revolver is still the cheapest in terms of interest and principal repayments.

AFM-09-14: Debt Schedule, Part 3: Dividend Recap (10:56)

In this lesson, you'll implement a Dividend Recap in Year 3 or 4 of the LBO model and change the formulas in the Debt Schedule and on the financial statements to support this feature.

Table of Contents:

- **1:03:** Debt Schedule
- **5:44:** Financial Statements
- **9:31:** Recap and Summary

Quiz Questions:

9:30 Why is a dividend recap in an LBO model unlikely to make a huge impact on the IRR to the private equity firm?

a) Because most dividend recaps are based on relatively low multiples of EBITDA.
b) Because a dividend recap only results in a partial recovery of the upfront investment earlier in the holding period.
c) Because a dividend recap also means additional Debt to repay in the holding period or upon exit.
d) All of the above.

Explanation: Everything above is true and explains why dividend recaps tend to make a small difference on the IRR and other output. They tend to be relatively small next to the upfront investment and exit value, and they also result in additional Debt that the company must repay, reducing its cash flow during the holding period and the exit equity value if the additional Debt is not fully repaid by then.

AFM-09-15: Tax Schedule and Net Operating Losses (NOL) in an Asset Purchase (9:09)

You will set up the Tax Schedule for the LBO model in this lesson and use it to track the company's NOL accruals and usages throughout the holding period.

Table of Contents:

- **1:20:** Taxable Income
- **3:35:** Net Operating Losses
- **6:34:** Linking the Statements
- **8:05:** Recap and Summary

Quiz Questions:

8:04 In which of the following LBO scenarios would a Tax and NOL Schedule be likely to make the BIGGEST difference?

a) The acquisition of a large, public company for a small premium.
b) The acquisition of a large, public company for a large premium.
c) A divestiture of one specific division at a huge premium to Net Assets.
d) The acquisition of a small, closely held private company for a high EBITDA multiple.

Explanation: The third answer choice is the correct one. Taxes make the biggest impact in an LBO model when items that are not normally cash-tax deductible, such as Goodwill Amortization, become deductible at significant levels, which is what happens when the deal is

structured as an asset purchase rather than a stock purchase (most common for divestitures and smaller, private companies). The first two answers are not correct because public company acquisitions are almost always stock purchases, and the last answer is not the best one because we don't know whether or not the equity purchase price represents a huge premium to the company's Net Assets (a high multiple increases the likelihood but doesn't necessarily prove it).

AFM-09-16: Credit Stats and Ratios, ROIC, and LBO Exit Multiples (16:13)

In this lesson, you will calculate the credit stats and ratios for Viridor and use its Return on Invested Capital (ROIC) before and after the deal to estimate the proper range of EBITDA exit multiples in the different scenarios.

Table of Contents:

- **1:25:** EBITDA, Debt, and Interest
- **6:16:** Debt Paydown, Growth, Margins, and ROIC
- **13:02:** Conclusions and Assumption Changes
- **15:06:** Recap and Summary

Quiz Questions:

15:05 Why might the ROIC calculation in an LBO model differ in the pre-transaction and post-transaction periods?

a) The tax rate often changes after the deal takes place.
b) There may be additional D&A on assets written up in the deal, which affects EBIT.
c) You might treat operating lease liabilities differently in each period.
d) All of the above.

Explanation: The second answer choice is correct. It makes no sense to assume a different tax rate or to treat operating lease liabilities after the deal closes (unless the tax rate is set to change in a future year), but there is frequently additional depreciation and amortization on asset write-ups in leveraged buyouts. That additional D&A is deducted to calculate EBIT or Operating Income, and NOPAT is based on EBIT * (1 – Tax Rate).

AFM-09-17: Exit Equity Value and Project-Level Returns (11:14)

You'll estimate the Exit Equity Value in this LBO model in different exit years in this lesson, factoring in the debt repayment, prepayment penalties, and equity grants to the mezzanine

investors, and you'll calculate the returns to the PE firm before the dividend recap and management earn-outs.

Table of Contents:

- **0:55:** Exit Equity Value & Debt Repayments
- **5:19:** Prepayment Penalties & Equity Grants
- **8:33:** Project-Level Returns
- **10:12:** Recap and Summary

Quiz Questions:

[No quiz questions.]

AFM-09-18: Management Earnouts and Sponsor Returns with Dividend Recap (8:21)

You'll estimate the equity returns, i.e., the returns to the private equity firm, in this lesson, factoring in the earn-outs paid to management, the dividend recap, the stub period, and the scenarios and variable exit years.

Table of Contents:

- **0:28:** Earn-Outs to Management
- **4:00:** Returns to Sponsor
- **7:30:** Recap and Summary

Quiz Questions:

7:29 In an LBO model, is a Dividend Recap likely to make a bigger impact on the PE firm's IRR or returns multiple?

a) **The IRR.**
b) The returns multiple.
c) It will make about the same percentage impact on both.

Explanation: A Dividend Recap will make a bigger impact on the IRR because of the time value of money. Essentially, it means that instead of waiting to get some of their invested capital back at the end of the deal (the exit), the PE firm can recover some of this upfront investment earlier in the holding period. This doesn't necessarily change the *multiple* by much (if anything), but it will boost the IRR in most cases since it shifts the returns to an earlier year.

AFM-09-19: Returns to Lenders (OID, Issuance Fees, Prepayment Penalties, and Equity Granted) (14:14)

In this lesson, you'll estimate the IRRs, multiples, and recovery percentages for the lenders, including the Senior Unsecured Noteholders, Subordinated Noteholders, and Mezzanine investors, including the impact of OID, issuance fees, cash interest and principal repayments, prepayment penalties, and equity granted.

Table of Contents:

- **1:13:** Senior Unsecured Notes
- **4:45:** Subordinated Notes
- **8:40:** Mezzanine
- **11:19:** Conclusions
- **13:04:** Recap and Summary

Quiz Questions:

13:03 How does Paid-in-Kind (PIK) Interest affect the IRR to lenders in an LBO?

a) It makes no direct impact on the cash flows, but it does indirectly affect components of the cash flows.
b) It increases the Cash Interest over time by increasing the Debt principal.
c) It increases the "Repayment Upon Exit" and "Prepayment Penalty" lines by increasing the Debt principal.
d) All of the above.

Explanation: Everything above is correct. PIK Interest does not appear as an explicit line item in the returns schedule for lenders, but it affects many line items in this schedule, including the Cash Interest, Principal Repayment Upon Exit, and Prepayment Penalty, because it increases the Debt balance over time.

AFM-09-20: Sensitivities for Equity Investors and Lenders (10:07)

You'll set up sensitivity tables for the IRRs and recoveries for the different investor groups in this lesson, and you'll draw quick conclusions about the conditions that might make the deal work more effectively.

Table of Contents:

- **1:15:** Creating the Tables and Determining the Ranges
- **6:23:** Conclusions and Fixes
- **9:19:** Recap and Summary

Quiz Questions:

9:18 Why might the IRR in an LBO be higher for both the lenders and the equity investors with an earlier exit date?

a) The exit multiple might not increase over time to compensate for the longer holding period in later years.
b) The time value of money makes it easier to earn higher IRRs based on the same money-on-money multiple if the holding period is shorter.
c) Items like dividends, OID, and prepayment penalties contribute more to the IRR when the holding period is shorter.
d) All of the above.

Explanation: Everything above is true. For the same money-on-money multiple, the IRR will always be higher in a shorter holding period because the annualized rate of return increases. One-time items, such as dividends, OID, and prepayment penalties, are also more significant for the IRR when the holding period is shorter because they're "distributed" over fewer years. The first answer choice makes the point that if the exit multiple remains at 20x (for example), and the company's EBITDA does not grow much, the time value of money means that the IRR is likely to be lower with a later exit date.

AFM-09-21: Investment Recommendation Presentation (Equity) (11:06)

In this lesson, you'll learn how to structure and draft a 10-slide investment recommendation presentation for this leveraged buyout, including both market data from outside research and the numerical output from the LBO model.

Table of Contents:

- **0:28:** Outline and Structure
- **1:26:** Overview
- **3:15:** Market
- **4:02:** LBO Model Output
- **8:04:** Counter-Factual
- **9:49:** Recap and Summary

Quiz Questions:

9:49 How might you use data on a company's market in an investment recommendation for a leveraged buyout?

a) To determine whether the revenue growth assumptions match the market growth rate.
b) To figure out if the company's spending and CapEx match historical and market norms.
c) To assess the potential for add-on acquisitions and further market consolidation.
d) All of the above.

Explanation: Everything above is true. You can use the market data to assess the industry's overall attractiveness, the company's ability to make add-on acquisitions, and the plausibility of the key revenue, expense, and CapEx assumptions.

AFM-09-22: Investment Recommendation Presentation for Lenders (10:06)

You'll learn how a 10-slide investment recommendation for the lenders in this deal might differ in this lesson, including how to present the numbers and market data differently and highlight different risk factors and possible mitigants.

Table of Contents:

- **0:38:** Outline and Structure
- **2:26:** Overview
- **4:06:** Market
- **4:37:** The Numbers
- **6:40:** Risk Factors
- **8:51:** Recap and Summary

Quiz Questions:

6:39 How might "The Numbers" section of an investment recommendation presentation differ for the equity investors vs. the lenders?

a) The lender version would focus on the range of outcomes in different scenarios.
b) The lender version would focus on the chances of losing money in different scenarios.
c) The lender version would not include references to the IRR output.
d) All of the above.

Explanation: The second answer choice is correct. The equity investors and the lenders care about their respective IRRs and the deal's outcome in different scenarios, but lenders pay far more attention to the downside cases because their upside is relatively capped (aside from special features such as equity granted to certain lender groups). This focus on downside risk is the key difference between equity and debt investment recommendations.